python indexing and slicing

In [1]:
```
s1 = 'welcometeam'
s1
```

Out[1]:
```
'welcometeam'
```

In [2]:
```
type(s1)
```

Out[2]:
```
str
```

In [3]:
```
len(s1)
```

Out[3]:
```
11
```

In [4]:
```
s1
```

Out[4]:
```
'welcometeam'
```

In [5]:
```
s1[1] # forward index
```

Out[5]:
```
'e'
```

In [6]:
```
s1
```

Out[6]:
```
'welcometeam'
```

In [7]:
```
s1[-1]
```

Out[7]:
```
'm'
```

In [8]:
```
s1
```

Out[8]:
```
'welcometeam'
```

In [9]:
```
s1[:]
```

Out[9]:
```
'welcometeam'
```

In [10]:
```
s1[7:11]
```

Out[10]:
```
'team'
```

In [11]:
```
s1
```

Out[11]:
```
'welcometeam'
```

In [12]:
```
s1[7:]
```

Out[12]:
```
'team'
```

In [13]:
```
s1[:]
```

Out[13]: 'welcometeam'

In [14]:
```
s1
```

Out[14]: 'welcometeam'

In [15]:
```
s1[2:10]
```

Out[15]: 'lcometea'

In [16]:
```
s1[:10]
```

Out[16]: 'welcometea'

In [17]:
```
s1
```

Out[17]: 'welcometeam'

In [18]:
```
s1[4:]
```

Out[18]: 'ometeam'

In [19]:
```
s1
```

Out[19]: 'welcometeam'

In [20]:
```
s1[12:]
```

Out[20]: ''

In [21]:
```
s1[13]  # there is no index 13
```

```
---------------------------------------------------------------------------
IndexError                                Traceback (most recent call last)
Cell In[21], line 1
----> 1 s1[13]

IndexError: string index out of range
```

In [22]:
```
s1[:13]
```

Out[22]: 'welcometeam'

-s1[left:right] -s1[left:right:step]

In [23]:
```
s1
```

Out[23]: 'welcometeam'

In [24]:
```
s1[:15]
```

Out[24]: 'welcometeam'

In [25]:
```
s1[1:8:2]
```

Out[25]: 'ecmt'

In [26]:
```
s1
```
Out[26]:
```
'welcometeam'
```

In [27]:
```
s1[::3]  # it will give 3 steps
```
Out[27]:
```
'wcea'
```

In [28]:
```
s1[-9:-1:3]
```
Out[28]:
```
'lme'
```

In [29]:
```
s1
```
Out[29]:
```
'welcometeam'
```

In [31]:
```
s1[::-1] # it will print reverse list
```
Out[31]:
```
'maetemoclew'
```

In [32]:
```
s1[::-2]
```
Out[32]:
```
'meeolw'
```

In [33]:
```
s1[::-3]
```
Out[33]:
```
'mtoe'
```

In [ ]:
```
#
```